Lab 6: Training pipelines and project proposal

Reading and Learning Outcomes

Reading: Reddi, Chapter 8 AI Training: the sections on training-system architecture, data loading, optimization, profiling, and §8.7 Optimization Techniques; additionally, Chapter 5 §5.5 Model Development and Chapter 13 §13.3 MLOps Key Components for experiment tracking and reproducibility.

After completing this lab, students should be able to:

- Restructure training into a reproducible YAML-driven pipeline.
- Use MLflow and PyTorch profiling to record experiments and identify bottlenecks.
- Analyze worker count, batch size, throughput, memory, optimizer, and scheduler trade-offs.
- Resume interrupted training and prepare a feasible individual project proposal.

1. [20] Take your best model from Lab 3. Restructure training into a script driven by a single YAML config object. Include seed, checkpoint saving, and MLflow experiment logging. Log params, metrics, artifacts.
2. [15] Profile one epoch with `torch.profiler`. Report share of time on data loading, forward pass, backward pass, optimizer step.
3. [15] Vary `num_workers` `[0,1,2,4,8]`. Find the bottleneck. Plot.
4. [15] Batch-size study: `[32,64,128,256,512]`. Record throughput (samples/s), peak memory, final accuracy after 5 epochs.
5. [15] Compare learning-rate schedules: StepLR, CosineAnnealingLR, ReduceLROnPlateau. Compare SGD+momentum and Adam under equal 10-epoch budgets.
6. [10] Implement gradient accumulation to emulate batch 256 with batch 64 and accumulation 4. Check training behaviour is similar.
7. [10] Interrupt training at epoch 3, resume from checkpoint. Confirm curves continue smoothly.
8. [15] **Individual project proposal (one page):** problem, dataset, success metric, deployment constraints (latency ≤ 50 ms, size ≤ 5 MB), planned optimization (pruning, distillation, or quantization), student name and ID. Commit.

Graded evidence: profile breakdown, MLflow runs, study tables, approved individual proposal.